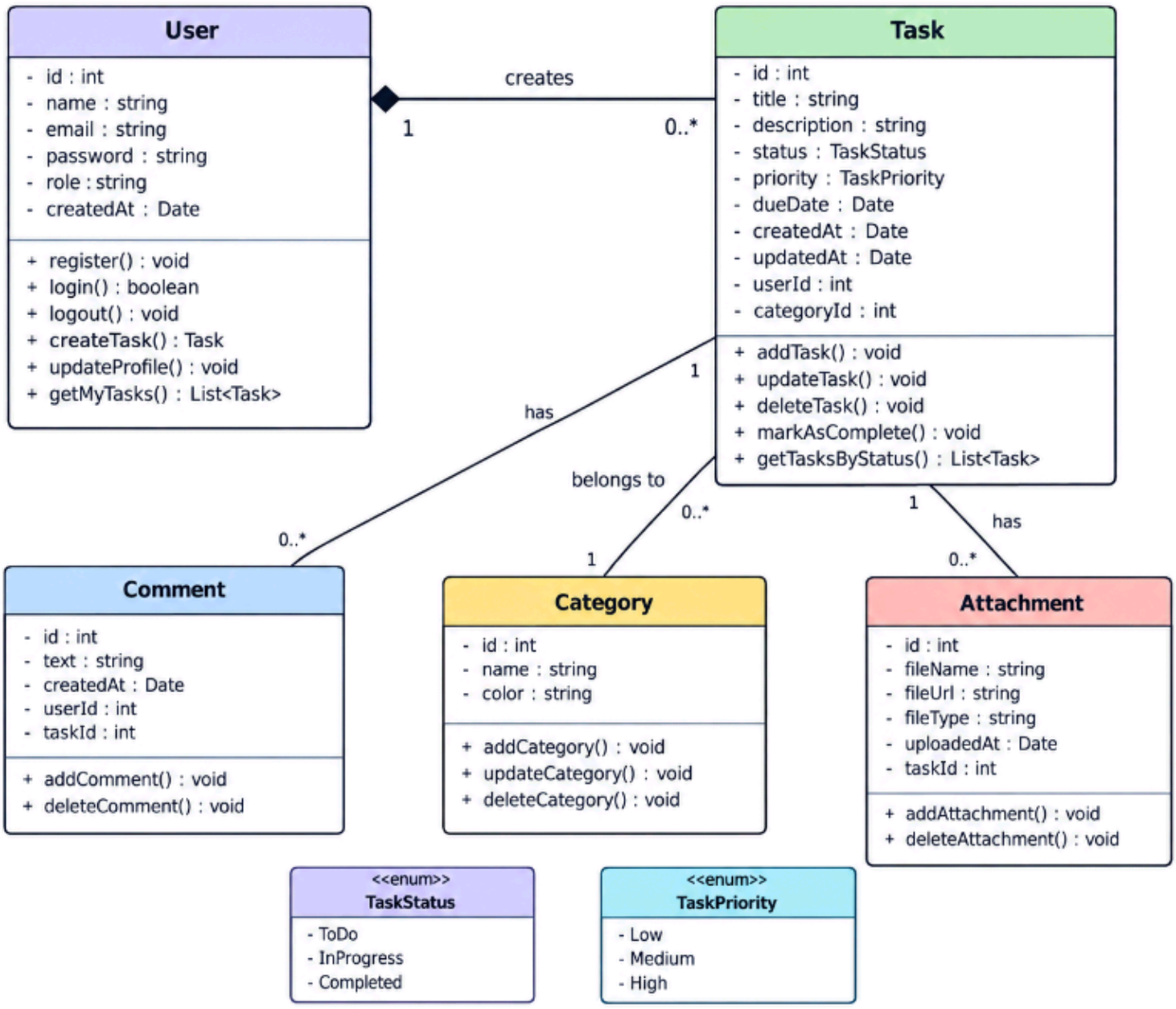
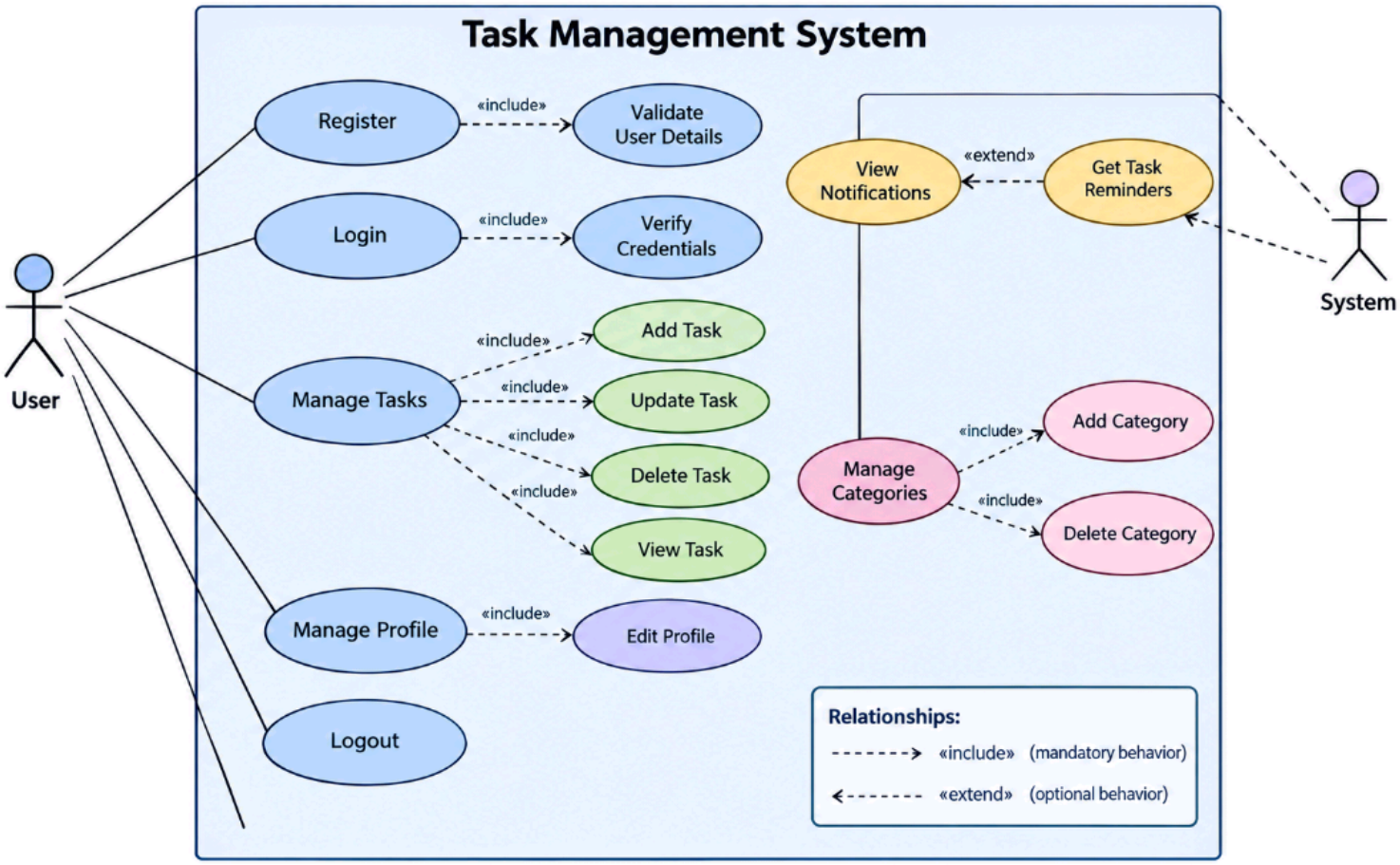
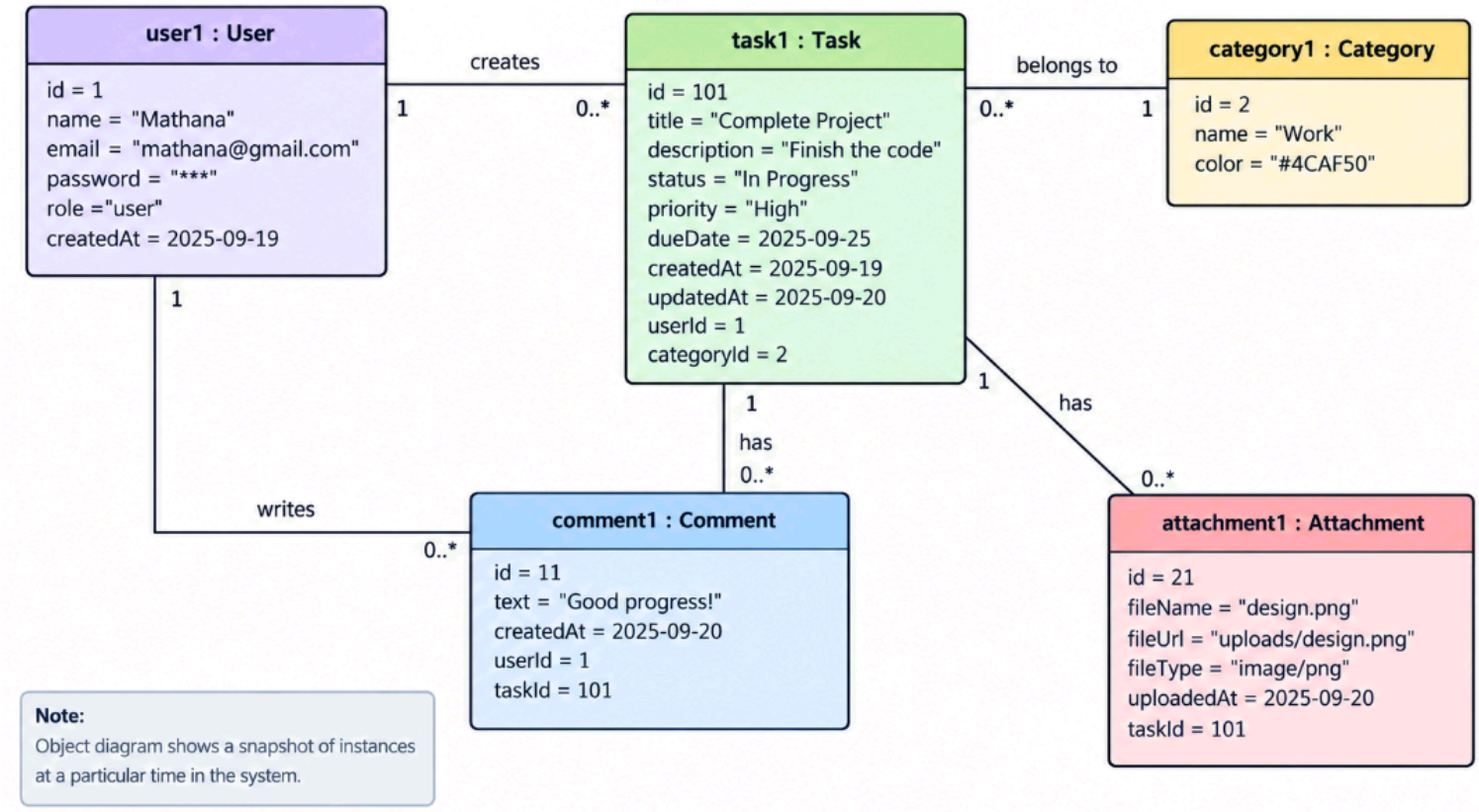
Task Management - Class Diagram

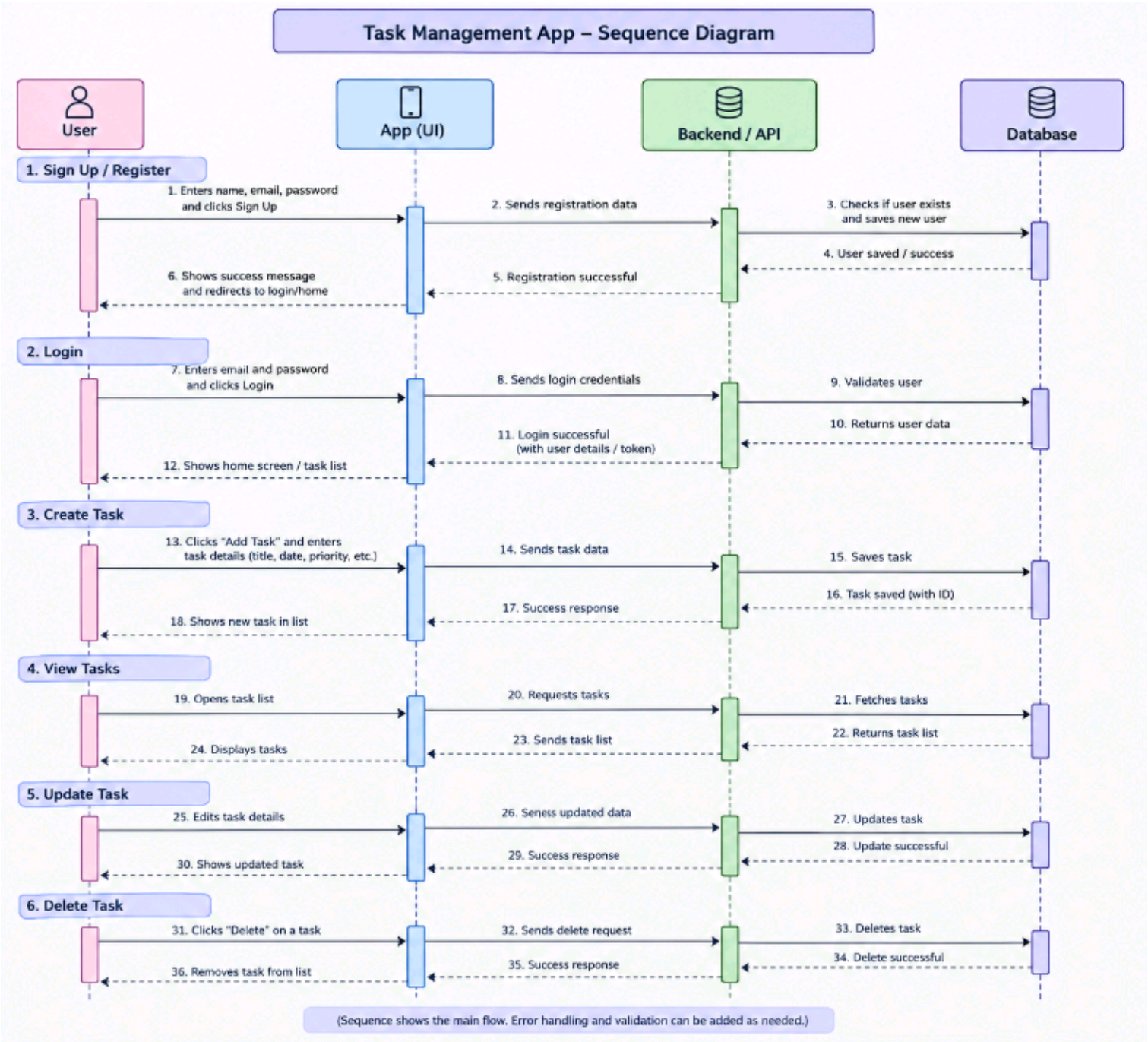



Task Management System
Register
«include»
Validate User Details
Login
«include»
Verify Credentials
Manage Tasks
«include»
Add Task
«include»
Update Task
«include»
Delete Task
«include»
View Task
Manage Profile
«include»
Edit Profile
Logout
View Notifications
«extend»
Get Task Reminders
Manage Categories
«include»
Add Category
«include»
Delete Category
User
System
Relationships:
«include» (mandatory behavior)
«extend» (optional behavior)

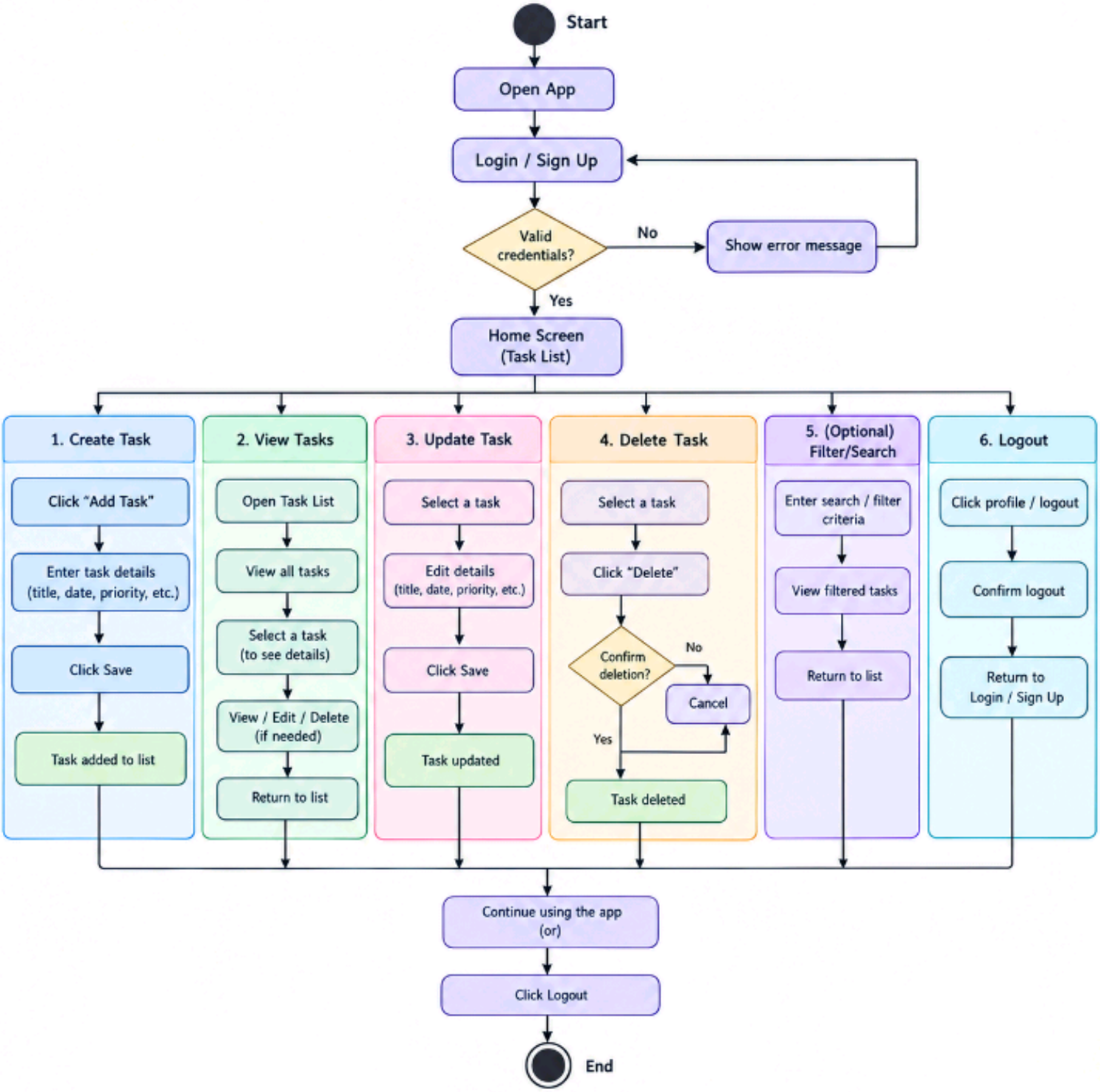
Task Management - Object Diagram

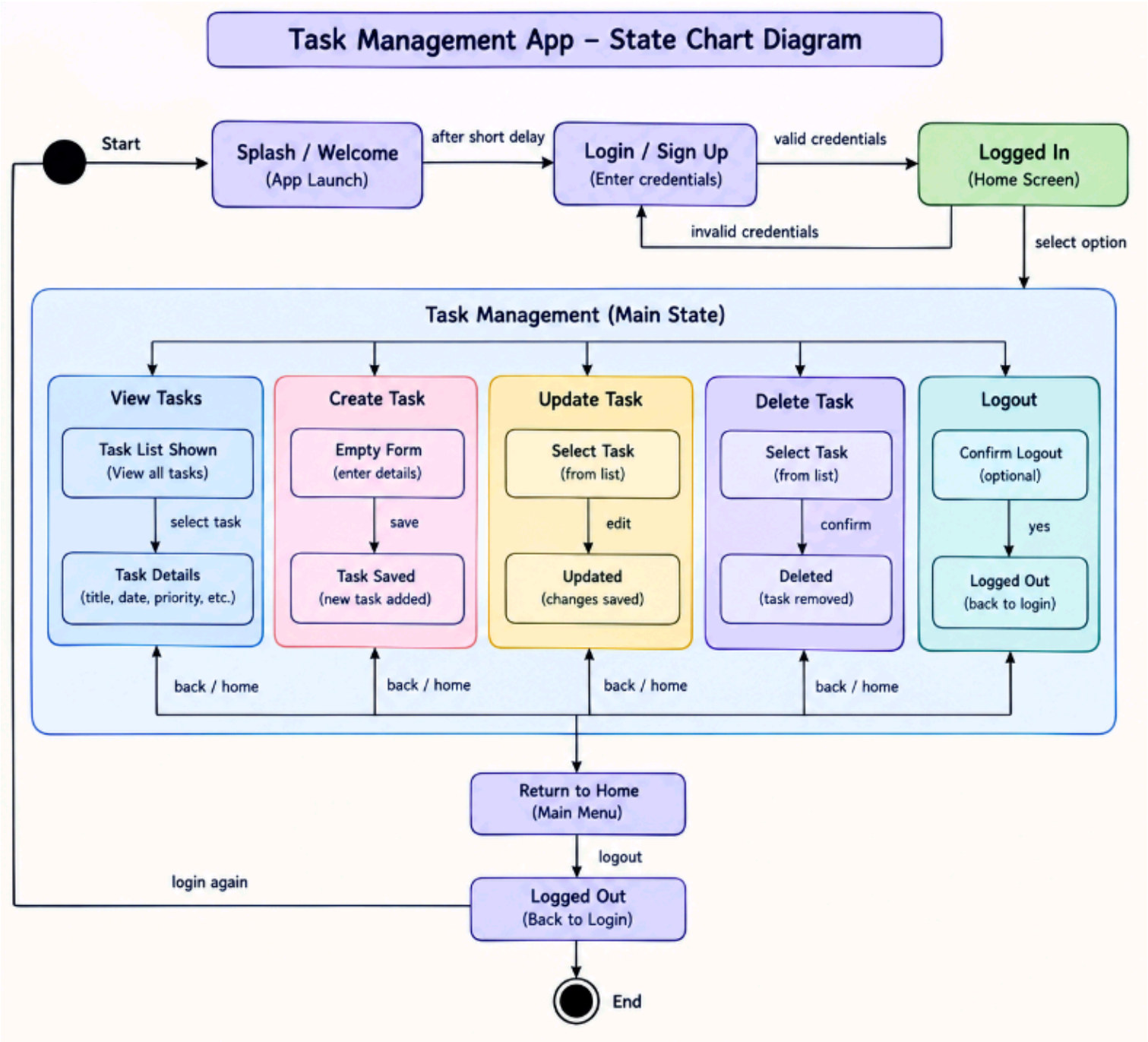


Note:
Object diagram shows a snapshot of instances at a particular time in the system.


Task Management App – Sequence Diagram
User
App (UI)
Backend / API
Database
1. Sign Up / Register
1. Enters name, email, password and clicks Sign Up
2. Sends registration data
3. Checks if user exists and saves new user
4. User saved / success
5. Registration successful
6. Shows success message and redirects to login/home
2. Login
7. Enters email and password and clicks Login
8. Sends login credentials
9. Validates user
10. Returns user data
11. Login successful (with user details / token)
12. Shows home screen / task list
3. Create Task
13. Clicks "Add Task" and enters task details (title, date, priority, etc.)
14. Sends task data
15. Saves task
16. Task saved (with ID)
17. Success response
18. Shows new task in list
4. View Tasks
19. Opens task list
20. Requests tasks
21. Fetches tasks
22. Returns task list
23. Sends task list
24. Displays tasks
5. Update Task
25. Edits task details
26. Seness updated data
27. Updates task
28. Update successful
29. Success response
30. Shows updated task
6. Delete Task
31. Clicks "Delete" on a task
32. Sends delete request
33. Deletes task
34. Delete successful
35. Success response
36. Removes task from list
(Sequence shows the main flow. Error handling and validation can be added as needed.)

Task Management App – Activity Diagram




Task Management App – State Chart Diagram
Start
Splash / Welcome
(App Launch)
after short delay
Login / Sign Up
(Enter credentials)
valid credentials
Logged In
(Home Screen)
invalid credentials
select option
Task Management (Main State)
View Tasks
Task List Shown
(View all tasks)
select task
Task Details
(title, date, priority, etc.)
Create Task
Empty Form
(enter details)
save
Task Saved
(new task added)
Update Task
Select Task
(from list)
edit
Updated
(changes saved)
Delete Task
Select Task
(from list)
confirm
Deleted
(task removed)
Logout
Confirm Logout
(optional)
yes
Logged Out
(back to login)
back / home
back / home
back / home
back / home
Return to Home
(Main Menu)
logout
login again
Logged Out
(Back to Login)
End